Finite Abelian Representations: The Zero-Weight Formula and a Direct Matrix Verification

Proof and reproducible verification strategy

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Abstract

For a finite abelian group acting diagonally, the coefficient of the monomial symmetric function m_τ in the sigma moment-generating function is the number of monomial basis vectors whose total character is trivial. This note proves the resulting simultaneous-congruence formula, derives the diagonal Molien average and finite constant-term form, and records a direct matrix test. The accompanying marimo notebook constructs the represented group, builds the full tensor product of symmetric powers, forms the Reynolds projector, and compares its rank with an exact lattice count and a numerical character average.

1 Setup and the represented matrix group

Let

$$A = C_{m_1} \times \cdots \times C_{m_r}, \qquad M = \widehat{A} \cong \mathbb{Z}/m_1\mathbb{Z} \oplus \cdots \oplus \mathbb{Z}/m_r\mathbb{Z}.$$

Every complex representation of A splits into one-dimensional characters. Choose a weight basis and write

$$V = \bigoplus_{j=1}^{d} \mathbb{C}_{w_j}, \qquad w_j = (w_{1j}, \dots, w_{rj}) \in M.$$

The weights are the columns of an integer matrix $W = (w_{\ell j}) \in \mathbb{Z}^{r\times d}$, understood row by row modulo m_ℓ. For $a = (a_1, \dots, a_r)$, the corresponding matrix is

$$\rho(a) = \operatorname{diag}_{1\le j\le d}\left(\exp\left(2\pi i \sum_{\ell=1}^{r} \frac{a_\ell w_{\ell j}}{m_\ell}\right)\right). \tag{1}$$

Thus the concrete matrix group is obtained by letting $0 \le a_\ell < m_\ell$ in (1). If ρ has a kernel, several abstract elements give the same matrix. These repetitions must be retained in an average over A.

For a partition or finite tuple $\tau = (\tau_1, \dots, \tau_s)$, put

$$T_\tau(V) = \bigotimes_{b=1}^{s} \operatorname{Sym}^{\tau_b} V.$$

The empty tuple gives $T_\varnothing(V) = \mathbb{C}$.

2 The simultaneous-congruence formula

Theorem 1 (Zero-weight formula)**.** *With the notation above,*

$$\dim T_\tau(V)^A = \#\left\{ (q_{bj}) \in \mathbb{Z}_{\geq 0}^{s\times d} : \begin{array}{ll} \displaystyle\sum_{j=1}^{d} q_{bj} = \tau_b & (1 \leq b \leq s), \\ \displaystyle\sum_{b=1}^{s}\sum_{j=1}^{d} w_{\ell j} q_{bj} \equiv 0 \pmod{m_\ell} & (1 \leq \ell \leq r) \end{array} \right\}. \tag{2}$$

Equivalently, for $q_b = (q_{b1}, \dots, q_{bd})^T$ *and* $D = \mathrm{diag}(m_1, \dots, m_r)$*,*

$$\dim T_\tau(V)^A = \#\Big\{ (q_1, \dots, q_s) : q_b \in \mathbb{Z}^d_{\geq 0}, \quad \mathbf{1}^T q_b = \tau_b, \quad W(q_1 + \cdots + q_s) \in D\mathbb{Z}^r \Big\}. \tag{3}$$

Proof. The monomials

$$e_1^{q_{b1}} \cdots e_d^{q_{bd}}, \qquad q_{bj} \geq 0, \quad \sum_j q_{bj} = \tau_b,$$

form a basis of $\mathrm{Sym}^{\tau_b} V$. Tensoring these bases over b gives a basis of $T_\tau(V)$ indexed by the arrays in the first line of (2). Since e_j has character w_j, the basis vector indexed by (q_{bj}) has total character

$$\sum_{b=1}^{s}\sum_{j=1}^{d} q_{bj} w_j \in M.$$

It is fixed by every element of A exactly when that character is zero. In coordinates this is precisely the second line of (2). Because the displayed monomial basis is a weight basis, the invariant subspace is spanned by exactly those basis vectors of weight zero. Counting them proves (2); collecting the q_b and writing divisibility by each m_ℓ as membership in $D\mathbb{Z}^r$ gives (3). □

3 Relation to the sigma-MGF and Molien averaging

Let $t_1, t_2, \dots$ be the alphabet used for the sigma-MGF. Inserting the diagonal eigenvalues of (1) into the generalized Molien average gives

$$\mathbb{E}_A[\mathrm{Exp}_\sigma(X_A h_1)] = \frac{1}{|A|} \sum_{a\in A} \prod_{i\geq 1} \prod_{j=1}^{d} \frac{1}{1 - t_i \chi_{w_j}(a)}. \tag{4}$$

For fixed, distinct alphabet indices, the coefficient of $t_{i_1}^{\tau_1} \cdots t_{i_s}^{\tau_s}$ in the product for a is

$$\prod_{b=1}^{s} h_{\tau_b}\big(\chi_{w_1}(a), \dots, \chi_{w_d}(a)\big) = \mathrm{tr}\big(T_\tau(\rho(a))\big).$$

The expression in (4) is symmetric in the t_i, so this is also the coefficient of m_τ. Therefore

$$[m_\tau]\, \mathbb{E}_A[\mathrm{Exp}_\sigma(X_A h_1)] = \frac{1}{|A|} \sum_{a\in A} \prod_{b=1}^{s} h_{\tau_b}\big(\chi_{w_1}(a), \dots, \chi_{w_d}(a)\big) = \dim T_\tau(V)^A. \tag{5}$$

The final equality follows from the Reynolds projector

$$P_A = \frac{1}{|A|} \sum_{a\in A} T_\tau(\rho(a)).$$

Indeed, P_A is the projection onto $T_\tau(V)^A$, hence $\operatorname{tr} P_A = \operatorname{rank} P_A = \dim T_\tau(V)^A$.

There is also a clean constant-term statement. Work in the group algebra with $z_\ell^{m_\ell} = 1$ and put $z^{w_j} = \prod_\ell z_\ell^{w_{\ell j}}$. If CT_A extracts the coefficient of the identity character, then

$$\mathbb{E}_A[\mathrm{Exp}_\sigma(X_A h_1)] = \mathrm{CT}_A \prod_{i \ge 1} \prod_{j=1}^{d} (1 - t_i z^{w_j})^{-1}, \tag{6}$$

$$[m_\tau]\, \mathbb{E}_A[\mathrm{Exp}_\sigma(X_A h_1)] = \mathrm{CT}_A \prod_{b=1}^{s} h_{\tau_b}(z^{w_1}, \ldots, z^{w_d}). \tag{7}$$

Character orthogonality identifies this operator with the finite Fourier projector

$$\mathrm{CT}_A\, F = \frac{1}{|A|} \sum_{a \in A} F\big(\zeta_{m_1}^{a_1}, \ldots, \zeta_{m_r}^{a_r}\big).$$

4 Verification strategy

The accompanying notebook uses three routes with different failure modes.

1. **Direct represented matrices.** It constructs all $|A| = \prod_\ell m_\ell$ matrices in (1). It first checks the identity, unitarity, and homomorphism laws numerically for every pair of represented elements. In the monomial basis, the diagonal of $\mathrm{Sym}^n(\rho(a))$ is indexed by weak compositions q of n and has entry $\prod_j \rho(a)_{jj}^{q_j}$. Kronecker products give the full matrices on $T_\tau(V)$. Their average is P_A; its numerical rank and trace give the target quantity directly. The residual $\|P_A^2 - P_A\|_F$ checks that the average is a projection.

2. **Exact zero-weight enumeration.** It enumerates the same weak compositions using integer arithmetic only and counts those for which every row of $W(q_1 + \cdots + q_s)$ is divisible by the corresponding m_ℓ. This is a literal implementation of (3); no complex roots of unity are used.

3. **Character/Molien average.** It computes
$$\frac{1}{|A|} \sum_{a \in A} \prod_b \operatorname{tr}\big(\mathrm{Sym}^{\tau_b}(\rho(a))\big)$$
in floating-point complex arithmetic. The result must be within 10^{-8} of the exact count. This route avoids the large Kronecker matrices while still testing the Fourier projection independently of the modular test.

A case passes when the projector rank, its rounded trace, the exact count, and the rounded Molien average agree, and both the numerical discrepancy and idempotence residual are below 10^{-8}.

5 Representative results and size choice

The default case is

$$A = C_3 \times C_4, \qquad W = \begin{pmatrix} 1 & 0 & 2 \\ 0 & 1 & 3 \end{pmatrix}, \qquad d = 3, \qquad \tau = (3, 2).$$

Here $|A| = 12$ and

$$\dim T_{(3,2)}(V) = \binom{3+3-1}{3} \binom{2+3-1}{2} = 10 \cdot 6 = 60.$$

The notebook constructs twelve 60×60 target matrices. It obtains

$$\operatorname{rank} P_A = 2, \qquad \text{zero-weight count} = 2, \qquad \text{Molien average} = 2 + 3.8 \times 10^{-15} i,$$

with maximum displayed numerical discrepancy 8.21×10^{-15}.

The automated suite broadens the coverage as follows.

Representation	$\lvert A\rvert$	d	largest τ	max. target dim.	checks
C_2, trivial plus sign	2	2	$(3,2)$	12	5
C_4, nonfaithful	4	3	$(2,1)$	18	4
$C_2 \times C_2$, all characters	4	4	$(2,1)$	40	4
$C_3 \times C_4$, mixed weights	12	3	$(3,2)$	60	5
$C_2 \times C_3$, repeated weights	6	4	$(2,2)$	100	4
				Total	22

All 22 checks pass. The cases include the empty partition, trivial weights, repeated weights, a nonfaithful representation, one and two cyclic factors, and tensor products with repeated and unequal degrees.

In general the target dimension is

$$N = \prod_{b=1}^{s} \binom{\tau_b + d - 1}{\tau_b}. \tag{8}$$

Dense direct projection requires $O(N^2)$ memory and approximately $O(|A|N^2)$ arithmetic, while literal congruence enumeration visits N monomial basis vectors. The notebook caps the dense target dimension at $N = 1000$ and uses $N = 60$ by default. For larger dimensions, the Molien trace average or an exact convolution of residue distributions on $\prod_\ell \mathbb{Z}/m_\ell\mathbb{Z}$ is the appropriate replacement for a dense projector.

6 Reproduction

From the repository root, launch the notebook with

```
.venv/bin/marimo edit \
  for_this_guy/finite_abelian_zero_weight_formula/finite_abelian_verification.py
```

The controls accept comma-separated moduli and partition parts. Rows of W are comma-separated and separated from one another by semicolons. For example, the default matrix is entered as

```
1, 0, 2; 0, 1, 3
```

The automated table is evaluated whenever the notebook runs.